

DOEUN KIM

Los Angeles, CA | (213) 348-0503 | doeunkim.cs@gmail.com | linkedin.com/in/doeun-kim/ | github.com/doeunyk

M.S. Computer Science candidate at USC with experience building AI-powered applications, conversational workflows, and scalable ML systems, focused on practical generative AI and user-centered AI experiences.

EDUCATION

University of Southern California	Los Angeles, CA
M.S. in Computer Science (GPA: 3.88 / 4.0)	December 2026
Ewha Womans University	Seoul, South Korea
B.S. in Computer Science and Engineering & B.S. in Content Convergence (GPA: 4.21 / 4.3; Top 4%)	February 2025
Honors: The National Scholarship for Science and Engineering (Merit-based, 2022-2024)	
Ewha Capstone Design Poster Session (Merit Award, 2024)	

WORK EXPERIENCE

RainbirdGEO	Seoul, South Korea
Software Engineer Intern (Full-stack)	May 2021 – July 2022
- Developed and iteratively refined a React Native mobile application with a Node.js/Express backend, incorporating usability-driven feedback from 100+ users and collaborating across teams in an Agile environment - Analyzed 2,000+ user survey responses and monitored 100+ application issues to identify usability and reliability bottlenecks, informing iterative product improvements that increased user satisfaction from 2.9 to 4.1	

RESEARCH EXPERIENCE

Laboratory for Machine Learning, Health and Biomedicine @ University of Southern California	Los Angeles, CA
Graduate Research Assistant (Advisor: Prof. Ruishan Liu)	November 2025 – Present
- Developed a deimmunization optimization pipeline using beam search to optimize peptide mutations for reduced immunogenicity while preserving binding affinity, reducing search overhead in multi-model evaluation workflows - Developed a microservice AI4Drug research platform integrating RabbitMQ for asynchronous ML inference orchestration and singleton-managed model services to support scalable computational drug discovery experiments	
AI Computing Platform Lab @ Ewha Womans University	Seoul, South Korea
Undergraduate Researcher (Advisor: Prof. Jaehyeong Sim)	April 2024 – February 2025
- Directed an undergraduate research team on <i>Layer-Adaptive Quantization of Diffusion Models using Fisher Information</i>, achieving 49.4% model size reduction and 7% FID improvement; presented results at the 2024 IEIE Symposium (poster session) - Conducted and presented a study on analyzing diffusion model inference mechanisms using XAI techniques (Integrated Gradients, GradSHAP, Occlusion) on ImageNet1000; served as team lead and speaker at the KCC 2024 XAI Workshop	

ACADEMIC PROJECTS

Multimodal Entrance Detection System	January 2026 – Present
- Developed a multimodal entrance detection pipeline integrating aerial and street-view imagery, building footprints, and GPS trajectories to identify valid entrances from 64 candidates using Kimi-based VLM inference and spatial overlay visualizations - Designed GPS-aware ranking calibration for reranking, improving entrance localization F1 from 0.30 to 0.59 and AUPR from 0.17 to 0.45	
PokerBench - Small LLMs for Poker Decision-Making	August 2025 – December 2025
- Built scalable training and evaluation pipelines for benchmarking SLMs against LLM baselines on poker decision-making tasks - Fine-tuned SLMs with QLoRA and analyzed efficiency-performance trade-offs against LLM baselines on large-scale poker decision-making datasets (~560K samples) using Action Accuracy (AA) and Exact Match (EM)	
Jeans - AI-Based Photo Editing and Sharing Service	December 2024 – February 2025
- Led a 5-member team to build production-ready AI features, including fine-tuned Whisper models and GPT-powered voice agents with tool APIs, prompt validation, and automated multi-step interaction workflows - Designed and deployed ML inference APIs with FastAPI and PyTorch, scaling on AWS EC2 with NGINX and CI/CD	

TECHNICAL SKILLS

AI/ML: PyTorch, Transformers, LLM fine-tuning (QLoRA), Whisper, BLIP, Scikit-learn, Keras, OpenCV
Programming Languages: Python, SQL, JavaScript/TypeScript, Java, C/C++, HTML/CSS
Frameworks/DBs: FastAPI, Django, Express, Sequelize, Spring, React, Gradio, MongoDB, MariaDB, SQLite, MySQL
Cloud/DevOps: AWS (EC2, ELB, Elastic Beanstalk), NGINX, Docker, Git, CI/CD (Github Actions)